

TECHNICAL REPORT

Dynamic modules –

Part 6-6: Design guide – Failure mode effect analysis for optical units of dynamic modules



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**Dynamic modules –
Part 6-6: Design guide – Failure mode effect analysis for optical units of dynamic
modules**

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INTERNATIONAL ELECTROTECHNICAL COMMISSION

DYNAMIC MODULES –

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for optical units of dynamic modules**

FOREWORD

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IEC 62343-6-6, which is a Technical Report, has been prepared by subcommittee 86C: Fibre optic systems and active devices, of IEC technical committee 86: Fibre optics.

This second edition cancels and replaces the first edition published in 2011. It constitutes a technical revision.

This edition includes the following significant technical changes with respect to the previous edition:

- a) addition of the LCOS based and the DLP based wavelength selective switch (WSS);

b) addition of the multicast optical switch module (MCOS).

The text of this technical report is based on the following documents:

DTR	Report on voting
86C/1396/DTR	86C/1421/RVC

Full information on the voting for the approval of this technical report can be found in the report on voting indicated in the above table.

This publication has been drafted in accordance with the ISO/IEC Directives, Part 2.

A list of all parts in the IEC 62343 series, published under the general title *Dynamic modules*, can be found on the IEC website.

The committee has decided that the contents of this publication will remain unchanged until the stability date indicated on the IEC web site under "<http://webstore.iec.ch>" in the data related to the specific publication. At this date, the publication will be

- reconfirmed,
- withdrawn,
- replaced by a revised edition, or
- amended.

A bilingual version of this publication may be issued at a later date.

DYNAMIC MODULES –

Part 6-6: Design guide – Failure mode effect analysis for optical units of dynamic modules

1 Scope

This part of IEC 62343, which is a Technical Report, describes failure mode effect analysis (FMEA) for optical units of dynamic modules. FMEA is one of the effective and useful analysis methods to determine the reliability evaluation test items and conditions that are defined in future reliability qualification documents.

In order to estimate the lifetime for a module, there is a typical procedure. The first step is to identify the dominant failure modes. The second step is to determine the acceleration tests according to these failure modes. The third step is to carry out the test. The fourth step is to estimate the acceleration factors. Finally, the fifth step is to calculate the lifetime of the dynamic module.

IEC 61300-2 (all parts) defines environment and mechanical tests. This Technical Report describes the dominant failure mode for dynamic modules and relevant tests from IEC 61300-2 (all parts).

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IEC 61300-2-1, *Fibre optic interconnecting devices and passive components – Basic test and measurement procedures – Part 2-1: Tests – Vibration (sinusoidal)*

IEC 61300-2-4, *Fibre optic interconnecting devices and passive components – Basic test and measurement procedures – Part 2-4: Tests – Fibre/cable retention*

IEC 61300-2-9, *Fibre optic interconnecting devices and passive components – Basic test and measurement procedures – Part 2-9: Tests – Shock*

IEC 61300-2-17, *Fibre optic interconnecting devices and passive components – Basic test and measurement procedures – Part 2-17: Tests – Cold*

IEC 61300-2-18, *Fibre optic interconnecting devices and passive components – Basic test and measurement procedures – Part 2-18: Tests – Dry heat – High temperature endurance*

IEC 61300-2-19, *Fibre optic interconnecting devices and passive components – Basic test and measurement procedures – Part 2-19: Tests – Damp heat (steady state)*

IEC 61300-2-22, *Fibre optic interconnecting devices and passive components – Basic test and measurement procedures – Part 2-22: Tests – Change of temperature*

IEC 61300-2-44, *Fibre optic interconnecting devices and passive components – Basic test and measurement procedures – Part 2-44: Tests – Flexing of the strain relief of fibre optic devices*

3 Terms and definitions

No terms and definitions are listed in this document.

ISO and IEC maintain terminological databases for use in standardization at the following addresses:

- IEC Electropedia: available at <http://www.electropedia.org/>
- ISO Online browsing platform: available at <http://www.iso.org/obp>

4 Consideration of types of dynamic modules

There are many types of dynamic modules: dynamic channel equalizer, tuneable optical chromatic dispersion compensator, dynamic gain tilt equalizer, wavelength selective switch, wavelength blocker, optical performance monitor, optical switch, and so on. The main feature of dynamic modules is to control their performances during operation. In order to achieve their features, many kinds of control mechanisms are used for dynamic modules; micro electro mechanical system (MEMS), stepping motor, electromagnet, thermo optics, magnet optics, electro optics, liquid crystal device (LCD), and so on.

Table 1 shows the first guidance of categorization of dynamic modules and their mode of evaluation. Dynamic modules without an electrical circuit board can be considered similar to passive optical components for purposes of evaluation. On the other hand, for dynamic modules with a control circuit board, it is necessary to give special consideration. There are mainly two types of internal design for dynamic modules: those for which it is easy to divide the constituting parts to consider the reliability, and those for which it is not easy to divide. It is necessary to consider how to evaluate according to these structures.

This document describes FMEA only for optical units for dynamic modules. It is necessary to evaluate whole dynamic modules including control circuit boards and firmware if used.

5 Typical failure points

In addition to control circuit boards and control of moving parts, a typical optical unit for a dynamic module consists of the following parts: optical element, outer package, fibre pigtails, optical semiconductor chips, and joint points of these elements. These elements have their own failure mode; for example, break for pigtails and displacement for joint points. Moreover, these elements may have their acceleration factor of degradation; for example, joint points fixed by adhesive are generally weak against high humidity. This failure mode analysis can be referred to FMEA for passive optical components (refer to IEC 62005-3).

There are special considerations for dynamic modules. The following are some examples:

- when using MEMS, operating shock and vibration test are necessary because MEMS are sensitive to mechanical shock and vibration;
- when temperature control is used, the temperature cycling test is recommended because temperature control functions generally produce thermal stress;
- The temperature cycling test can accelerate thermal stress.

6 Failure modes and known failure mechanisms

For some dynamic modules, FMEA is carried out. Table 2 shows known failure mechanisms, failure effects, failure modes, relevant tests, and IEC test document numbers for dynamic modules. If new technology and new dynamic modules become commercially available, they will be added to Table 2 in later revisions of this document. Relevant tests are listed with the failure effect and the dominant failure mechanism. As other relevant tests or methods of failure mode excitation become known, these will also be added.

Table 1 – Categorization based on the structure and mode of evaluation

Electrical circuits		How to evaluate	Examples
Without electrical circuits	N.A.	As optical component	VOA, 1x2/2x2 optical switch, DGTE
With electrical circuit	Easy to divide optical and electrical unit	As optical and electrical units individually, and as integrated dynamic module	VOA, VOA-MUX, TDC (DCDC), DCE, matrix switch, channel monitor, performance monitor
	Difficult to divide optical and electrical unit	To evaluate as integrated dynamic module	Wavelength blocker (WB), Wavelength selective switch (WSS)
Optical active and passive components should comply with the reliability qualification requirement defined in IEC 62572 (all parts) for active components and IEC 62005-9 (all parts) for passive optical components, respectively. In cases in which it is difficult to divide optical and electrical units, integrated modules should be tested. Electrical circuit boards should be qualified individually. The following standards series are useful references for the quality of electrical circuit boards: IEC 61188 (all parts), IEC 61189 (all parts), IEC 61190 (all parts), and IEC 61191 (all parts). Three pieces should be tested.			

Table 2 – Failure mode and known failure mechanisms for the optical units of dynamic devices

Dynamic devices		Constitution parts	Known failure mechanisms	Failure modes	Degradation acceleration factors	Relevant tests	IEC references
Variable optical attenuators	MEMS type	MEMS	Stacking the moving part	Uncontrollable	Mechanical stress Excess driving power	Shock (storage) Vibration (storage) Maximum absolute rating test (electrical) On/off driving test	IEC 61300-2-9 IEC 61300-2-1 Under study Under study
			Distortion of hinge/mirror	Insertion loss increase Attenuation change Return loss decrease Dynamic range of attenuation decrease PDL increase WDL increase	Mechanical stress Thermal stress Excess driving power	Shock (storage) Vibration (storage) Shock and vibration (operating) Change of temperature Maximum absolute rating test On/off driving test	IEC 61300-2-9 IEC 61300-2-1 Under study IEC 61300-2-22 Under study Under study
			Reflectance of mirror changing	Insertion loss increase Attenuation change Return loss decrease PDL increase WDL increase	High humidity (non-hermetic sealed)	Damp heat	IEC 61300-2-19
		Collimator	Dislocation of fixing points of optical parts	Insertion loss increase Attenuation change Dynamic range of attenuation decrease PDL increase WDL increase	Thermal stress High humidity (non-hermetic sealed and using adhesive) Mechanical stress	Change of temperature High temperature Damp heat Shock (storage) Vibration (storage)	IEC 61300-2-22 IEC 61300-2-18 IEC 61300-2-19 IEC 61300-2-9 IEC 61300-2-1
		Pigtail	Fibre broken, micro-bending	Insertion loss increase No operation	Mechanical stress for pigtail	Fibre cable retention Optical fibre cable flexing	IEC 61300-2-4 IEC 61300-2-44

Dynamic devices		Constitution parts	Known failure mechanisms	Failure modes	Degradation acceleration factors	Relevant tests	IEC references
	Liquid crystal type	LCD	Degradation of LCD	Insertion loss increase	Thermal stress	Change of temperature	IEC 61300-2-22
				Attenuation change	High humidity (non-hermetic sealed)	High temperature	IEC 61300-2-18
				Return loss decrease	Mechanical stress	Damp heat	IEC 61300-2-19
		Electrical polarization of LCD	Uncontrollable	Excess driving power	Maximum absolute rating test	Shock (storage)	IEC 61300-2-9
						Vibration (storage)	IEC 61300-2-1
	Magnet optic type	Freezing of LCD	Uncontrollable	Low temperature	Cold		IEC 61300-2-17
		Collimator	Same as MEMS type				
		Pigtail	Same as MEMS type				
		Magnet optic part	Dislocation of magnet, Faraday rotator and birefringent crystal	Insertion loss increase	Thermal stress	Change of temperature	IEC 61300-2-22
				Attenuation change	High humidity (non-hermetic sealed and using adhesive)	High temperature	IEC 61300-2-18
				Return loss decrease	Mechanical stress	Damp heat	IEC 61300-2-19
	Mechanical type	Moving part	Stacking the moving part	Uncontrollable	Mechanical stress High humidity (non-hermetic) Excess driving power	Shock (storage)	IEC 61300-2-9
						Vibration (storage)	IEC 61300-2-1
						Damp heat	IEC 61300-2-19
	Collimator	Same as MEMS type	Same as MEMS type	Same as MEMS type	Maximum absolute rating test (electrical)	Under study	Under study
						On/off driving test	

Dynamic devices		Constitution parts	Known failure mechanisms	Failure modes	Degradation acceleration factors	Relevant tests	IEC references
			Degradation of moving part	Driving power increase	Mechanical stress Thermal stress Excess driving power High humidity (non-hermetic sealed)	Shock (storage) Vibration (storage) Shock and vibration (operating) Change of temperature Maximum absolute rating test On/off driving test Damp heat	IEC 61300-2-9 IEC 61300-2-1 Under study IEC 61300-2-22 Under study Under study IEC 61300-2-19
			Distortion of mirror	Insertion loss increase Return loss decrease Crosstalk increase PDL increase	Mechanical stress Thermal stress Excess driving power	Shock (storage) Vibration (storage) Change of temperature Maximum absolute rating test On/off driving test	IEC 61300-2-9 IEC 61300-2-1 IEC 61300-2-22 Under study Under study
			Reflectance of mirror changing	Insertion loss increase Attenuation change Return loss decrease PDL increase WDL increase	High humidity (non-hermetic sealed)	Damp heat	IEC 61300-2-19
		Collimator	Same as MEMS type				
		Pigtail	Same as MEMS type				
	Planar waveguide type (thermal optic effect)	Waveguide	Refractive index changing	Insertion loss increase Attenuation change Return loss decrease Dynamic range of attenuation decrease PDL increase	Thermal stress High humidity (non-hermetic sealed and using adhesive) Mechanical stress	Change of temperature High temperature Damp heat Shock (storage) Vibration (storage)	IEC 61300-2-22 IEC 61300-2-18 IEC 61300-2-19 IEC 61300-2-9 IEC 61300-2-1

Dynamic devices		Constitution parts	Known failure mechanisms	Failure modes	Degradation acceleration factors	Relevant tests	IEC references
			Electrode degradation	Dynamic range of attenuation decrease	High humidity (non-hermetic sealed and using adhesive) Excess driving power	Damp heat Maximum absolute rating test Diving test	IEC 61300-2-19 Under study Under study
		Fixing point between waveguide and fibres	Dislocation by the degradation of adhesive	Insertion loss increase	Thermal stress High humidity (non-hermetic sealed and using adhesive)	Change of temperature Damp heat	IEC 61300-2-22 IEC 61300-2-19
		Pigtail	Same as MEMS type				
Optical switches	Mechanical type	Moving part	Stacking the moving part	Uncontrollable	Same as VOA mechanical type		
			Degradation of moving part	Driving power increase Switching time increase	Same as VOA mechanical type		
			Distortion of mirror	Insertion loss increase Return loss decrease Crosstalk increase PDL increase	Same as VOA mechanical type		
			Reflectance of mirror changing	Insertion loss increase Return loss decrease PDL increase WDL increase	Same as VOA mechanical type		
		Collimator	Same as VOA MEMS type	Insertion loss increase Crosstalk increase PDL increase WDL increase	Same as VOA mechanical type		
		Pigtail	Same as VOA MEMS type				
	Planar waveguide type (thermal optic effect)	Waveguide	Refractive index changing	Insertion loss increase Crosstalk increase Return loss decrease PDL increase	Same as VOA PLC TO type		

Dynamic devices		Constitution parts	Known failure mechanisms	Failure modes	Degradation acceleration factors	Relevant tests	IEC references
			Electrode degradation	Crosstalk increase	Same as VOA PLC TO type		
		Fixing point between waveguide and fibres	Dislocation by the degradation of adhesive	Insertion loss increase	Same as VOA PLC TO type		
		Pigtail	Same as VOA MEMS type				
	MEMS type	MEMS	Stacking the moving part	Uncontrollable	Same as VOA MEMS type		
			Distortion of hinge/mirror	Insertion loss increase	Same as VOA MEMS type		
				Crosstalk increase Return loss decrease PDL increase WDL increase			
		Reflectance of mirror changing	Insertion loss increase Crosstalk increase Return loss decrease PDL increase WDL increase	Same as VOA MEMS type			
		Collimator	Same as VOA MEMS type	Same as VOA MEMS type except: Crosstalk increase instead of attenuation change	Same as VOA MEMS type		
	Pigtail	Same as VOA MEMS type					

Dynamic devices		Constitution parts	Known failure mechanisms	Failure modes	Degradation acceleration factors	Relevant tests	IEC references
	Electrical optics type (LN)	Waveguide	Refractive index changing	Insertion loss increase	Thermal stress	Change of temperature	IEC 61300-2-22
				Crosstalk increase	Mechanical stress	High temperature	IEC 61300-2-18
		Fixing point between waveguide and fibres	Dislocation by the degradation of adhesive	Driving voltage changing	Excess driving power	Damp heat	IEC 61300-2-19
						Shock (storage)	IEC 61300-2-9
						Vibration (storage)	IEC 61300-2-1
						Maximum absolute rating test	Under study
						Diving test	Under study
		Electrode degradation		Crosstalk increase	Same as switch PLC TO (VOA PLC TO) type		
		Pigtail	Same as VOA MEMS type				
Tuneable filters (electrical control)	MEMS etalon type	MEMS	Stacking the moving part	Uncontrollable	Same as VOA MEMS type		
			Distortion of hinge/mirror	Insertion loss increase	Same as VOA MEMS type		
				Centre-wavelength change			
				Return loss decrease			
			PDL increase				
			FWHM increase				
			Tuneable wavelength range decrease				
			Reflectance of etalon mirror changing	Insertion loss increase	Same as VOA MEMS type		
				Centre-wavelength change			
				Return loss decrease			
				PDL increase			
				FWHM increase			

Dynamic devices		Constitution parts	Known failure mechanisms	Failure modes	Degradation acceleration factors	Relevant tests	IEC references
		Collimator	Same as VOA MEMS type	Insertion loss increase Centre-wavelength change Return loss decrease PDL increase FWHM increase Tuneable wavelength range decrease	Same as VOA MEMS type		
		Pigtail	Same as VOA MEMS type				
	PZT etalon type	PZT	PZT degradation	Driving voltage increase	High humidity Thermal stress Excess driving	Damp heat Change of temperature Maximum absolute rating test On/off driving test	IEC 61300-2-19 IEC 61300-2-22 Under study Under study
		Etalon mirror	Reflectance of etalon mirror changing	Insertion loss increase Centre-wavelength change Return loss decrease PDL increase FWHM increase	Same as VOA MEMS type		
		Optical part and collimator	Dislocation of fixing points of optical parts	Insertion loss increase Centre-wavelength change FWHM increase Tuneable wavelength range decrease	Thermal stress High humidity (non-hermetic sealed and using adhesive) Mechanical stress	Change of temperature High temperature Damp heat Shock (storage) Vibration (storage)	IEC 61300-2-22 IEC 61300-2-18 IEC 61300-2-19 IEC 61300-2-9 IEC 61300-2-1
		Pigtail	Same as VOA MEMS type				
		Thin film filter mechanical	Moving part	Stacking the moving part	Uncontrollable	Same as VOA mechanical type	
			Degradation of moving part	Driving power increase	Same as VOA mechanical type		

Dynamic devices		Constitution parts	Known failure mechanisms	Failure modes	Degradation acceleration factors	Relevant tests	IEC references
	control type		Thin film filter degradation	Insertion loss increase Centre-wavelength change FWHM increase	High humidity (non-hermetic sealed)	Damp heat	IEC 61300-2-19
		Collimator	Same as VOA MEMS type	Insertion loss increase Centre-wavelength change Return loss decrease PDL increase FWHM increase Tuneable wavelength range decrease	Same as VOA MEMS type		
		Pigtail	Same as VOA MEMS type				
Dynamic chromatic dispersion compensators	VIPA type	Moving part	Degradation of stepping motor	Uncontrollable Dynamic range of CD decrease	Thermal stress Mechanical stress Excess driving High humidity (non-hermetic)	Shock (storage) Vibration (storage) Shock and vibration (operating) Change of temperature Maximum absolute rating test On/off driving test Damp heat	IEC 61300-2-9 IEC 61300-2-1 Under study IEC 61300-2-22 Under study Under study IEC 61300-2-19
		VIPA mirror	Distortion of VIPA mirror	Centre-wavelength change Insertion loss increase GDR increase	Thermal stress High humidity (non-hermetic) Mechanical stress	Change of temperature Damp heat High temperature Shock (storage) Vibration (storage)	IEC 61300-2-22 IEC 61300-2-19 IEC 61300-2-18 IEC 61300-2-9 IEC 61300-2-1

Dynamic devices		Constitution parts	Known failure mechanisms	Failure modes	Degradation acceleration factors	Relevant tests	IEC references
			Temperature uncontrollable	Centre-wavelength change Insertion loss increase	Thermal stress High humidity (non-hermetic) Mechanical stress	Change of temperature Damp heat High temperature Shock (storage) Vibration (storage)	IEC 61300-2-22 IEC 61300-2-19 IEC 61300-2-18 IEC 61300-2-9 IEC 61300-2-1
		Optical parts and collimator	Dislocation of fixing points of optical parts (3D mirror distortion)	Insertion loss increase Centre-wavelength change Dynamic range of DC decrease GDR increase	Thermal stress High humidity (non-hermetic sealed and using adhesive) Mechanical stress	Change of temperature Damp heat High temperature Shock (storage) Vibration (storage)	IEC 61300-2-22 IEC 61300-2-19 IEC 61300-2-18 IEC 61300-2-9 IEC 61300-2-1
		Pigtail	Same as VOA MEMS type				
	FBG thermal control type	Thermal control part	TEC/heater degradation	Insertion loss increase Wavelength range change Dynamic range of CD decrease	Thermal stress High humidity (non-hermetic) Mechanical stress Excess driving	Change of temperature Damp heat High temperature Shock (storage) Vibration (storage) Maximum absolute rating test On/off driving test	IEC 61300-2-22 IEC 61300-2-19 IEC 61300-2-18 IEC 61300-2-9 IEC 61300-2-1 Under study Under study
		Fixing point between FBG and temperature control	Fibre bending	Insertion loss increase Wavelength range change CD changing GDR increase	Thermal stress High humidity (non-hermetic) Mechanical stress	Change of temperature Damp heat Shock (storage) Vibration (storage)	IEC 61300-2-22 IEC 61300-2-19 IEC 61300-2-9 IEC 61300-2-1
		FBG	Refractive index changing	CD changing	High temperature	High temperature	IEC 61300-2-18
		Pigtail	Same as VOA MEMS type				

Dynamic devices		Constitution parts	Known failure mechanisms	Failure modes	Degradation acceleration factors	Relevant tests	IEC references
Dynamic gain tilt equalizers	Magnet optic type	Magnet optic part	Dislocation of magnet, Faraday rotator and birefringent crystal	Insertion loss increase Tilt change Return loss decrease PDL increase WDL increase	Same as VOA MO type		
		Collimator	Same as VOA MEMS type	Insertion loss increase Tilt change Return loss decrease PDL increase WDL increase	Same as VOA MEMS type		
		Pigtail	Same as VOA MEMS type				
	Planar waveguide type	Waveguide	Refractive index changing	Insertion loss increase Tilt change Return loss decrease PDL increase WDL increase	Same as VOA PLC TO type		
		Fixing point between waveguide and fibres	Dislocation by the degradation of adhesive	Insertion loss increase	Same as VOA PLC TO type		
		Pigtail	Same as VOA MEMS type				
		MEMS and diffractive grating type	MEMS	Stacking the moving part	Uncontrollable	Same as VOA MEMS type	
	Distortion of hinge/mirror			Insertion loss increase Attenuation change Return loss decrease Dynamic range of attenuation decrease PDL increase WDL increase	Same as VOA MEMS type		

Dynamic devices		Constitution parts	Known failure mechanisms	Failure modes	Degradation acceleration factors	Relevant tests	IEC references
			Reflectance of mirror changing	Insertion loss increase Attenuation change Return loss decrease PDL increase WDL increase	Same as VOA MEMS type		
		Diffraction grating, lens, prism and collimator	Dislocation of fixing points of optical parts	Insertion loss increase Attenuation change Return loss decrease Dynamic range of attenuation decrease PDL increase WDL increase	Thermal stress High humidity (non-hermetic sealed and using adhesive) Mechanical stress	Change of temperature High temperature Damp heat Shock (storage) Vibration (storage)	IEC 61300-2-22 IEC 61300-2-18 IEC 61300-2-19 IEC 61300-2-9 IEC 61300-2-1
		Pigtail	Same as VOA MEMS type				
	Liquid crystal and diffraction grating type	LCD	Degradation of LCD	Insertion loss increase Attenuation change Return loss decrease Dynamic range of attenuation decrease PDL increase WDL increase	Same as VOA LCD type		
			Electrical polarization of LCD	Uncontrollable	Same as VOA LCD type		
			Freezing of LCD	Uncontrollable	Same as VOA LCD type		

Dynamic devices		Constitution parts	Known failure mechanisms	Failure modes	Degradation acceleration factors	Relevant tests	IEC references
		Diffraction grating, lens, prism and collimator	Dislocation of fixing points of optical parts	Insertion loss increase Attenuation change Return loss decrease Dynamic range of attenuation decrease PDL increase WDL increase	Same as DCE MEMS type		
		Pigtail	Same as VOA MEMS type				
Wavelength blockers	MEMS and diffraction grating type	MEMS	Stacking the moving part	Uncontrollable	Same as VOA MEMS type		
			Distortion of hinge/mirror	Insertion loss increase Attenuation change Return loss decrease Dynamic range of attenuation decrease PDL increase WDL increase	Same as VOA MEMS type		
			Reflectance of mirror changing	Insertion loss increase Attenuation change Return loss decrease PDL increase WDL increase	Same as VOA MEMS type		

Dynamic devices		Constitution parts	Known failure mechanisms	Failure modes	Degradation acceleration factors	Relevant tests	IEC references
		Diffractive grating, lens, prism and collimator	Dislocation of fixing points of optical parts	Insertion loss increase Attenuation change Return loss decrease Dynamic range of attenuation decrease PDL increase WDL increase	Same as DCE MEMS type		
		Pigtail	Same as VOA MEMS type				
	Liquid crystal and diffractive grating type	LCD	Degradation of LCD	Insertion loss increase Attenuation change Return loss decrease Dynamic range of attenuation decrease PDL increase WDL increase	Same as VOA LCD type		
				Electrical polarization of LCD	Uncontrollable	Same as VOA LCD type	
				Freezing of LCD	Uncontrollable	Same as VOA LCD type	
		Diffractive grating, lens, prism and collimator	Dislocation of fixing points of optical parts	Insertion loss increase Attenuation change Return loss decrease Dynamic range of attenuation decrease PDL increase WDL increase	Same as DCE MEMS type		
					Pigtail	Same as VOA MEMS type	

Dynamic devices		Constitution parts	Known failure mechanisms	Failure modes	Degradation acceleration factors	Relevant tests	IEC references
Wavelength selectable switches	MEMS and diffractive grating type	MEMS	Stacking the moving part	Uncontrollable	Same as VOA MEMS type		
			Distortion of hinge/mirror	Insertion loss increase	Same as VOA MEMS type		
		Attenuation change					
		Return loss decrease					
			Dynamic range of attenuation decrease				
	PDL increase						
	WDL increase						
	Diffractive grating, lens, prism and collimator	Reflectance of mirror changing	Insertion loss increase	Same as VOA MEMS type			
Attenuation change							
Return loss decrease							
		PDL increase					
		WDL increase					
	Diffractive grating, lens, prism and collimator	Dislocation of fixing points of optical parts	Insertion loss increase	Same as DCE MEMS type			
Attenuation change							
		Return loss decrease					
		Dynamic range of attenuation decrease					
		PDL increase					
		WDL increase					
	Pigtail	Same as VOA MEMS type					
	DLP and diffractive grating type	DLP	Stacking the moving parts	Insertion loss increase	Same as WSS MEMS type		
Attenuation change							
Return loss decrease							
Dynamic range of attenuation decrease							

Dynamic devices		Constitution parts	Known failure mechanisms	Failure modes	Degradation acceleration factors	Relevant tests	IEC references
			Reflectance of mirror changing	Insertion loss increase Attenuation change Return loss decrease PDL increase WDL increase	Same as WSS MEMS type		
		Diffraction grating, lens, prism and collimator	Dislocation of fixing points of optical parts	Insertion loss increase Attenuation change Return loss decrease Dynamic range of attenuation decrease PDL increase WDL increase	Same as WSS MEMS type		
		Pigtail	Same as VOA MEMS type				
	Liquid crystal and diffraction grating type	LCD	Degradation of LCD	Insertion loss increase Attenuation change Return loss decrease Dynamic range of attenuation decrease PDL increase WDL increase	Same as VOA LCD type		
			Electrical polarization of LCD	Uncontrollable	Same as VOA LCD type		
			Freezing of LCD	Uncontrollable	Same as VOA LCD type		

Dynamic devices		Constitution parts	Known failure mechanisms	Failure modes	Degradation acceleration factors	Relevant tests	IEC references			
		Diffraction grating, lens, prism and collimator	Dislocation of fixing points of optical parts	Insertion loss increase Attenuation change Return loss decrease Dynamic range of attenuation decrease PDL increase WDL increase	Same as DCE MEMS type					
		Pigtail	Same as VOA MEMS type							
	LCOS and diffraction grating type	LCOS	Degradation of LCD	Insertion loss increase Attenuation change Return loss decrease Dynamic range of attenuation decrease PDL increase WDL increase	Same as WSS LCD type					
				Electrical polarization of LCD				Uncontrollable	Same as WSS LCD type	
				Freezing of LCD				Uncontrollable	Same as WSS LCD type	
		Diffraction grating, lens, prism and collimator	Dislocation of fixing points of optical parts	Insertion loss increase Attenuation change Return loss decrease Dynamic range of attenuation decrease PDL increase WDL increase	Same as DCE MEMS type					
	Pigtail	Same as VOA MEMS type								

Dynamic devices		Constitution parts	Known failure mechanisms	Failure modes	Degradation acceleration factors	Relevant tests	IEC references
	Planar waveguide type (thermal optic effect)	Waveguide	Refractive index changing	Insertion loss increase	Same as VOA PLC TO type		
				Crosstalk increase			
		Return loss decrease					
		PDL increase					
			Electrode degradation	Crosstalk increase	Same as VOA PLC TO type		
		Fixing point between waveguide and fibres	Dislocation by the degradation of adhesive	Insertion loss increase	Same as VOA PLC TO type		
		Pigtail	Same as VOA MEMS type				
	Planar waveguide and MEMS type	MEMS	Stacking the moving part	Uncontrollable	Same as VOA MEMS type		
			Distortion of hinge/mirror	Insertion loss increase	Same as VOA MEMS type		
				Isolation decrease			
		Return loss decrease					
		PDL increase					
	WDL increase						
	Reflectance of mirror changing	Insertion loss increase	Same as VOA MEMS type				
		Isolation decrease					
		Return loss decrease					
		PDL increase					
		WDL increase					
	Waveguide	Refractive index changing	Insertion loss increase	Same as VOA PLC TO type			
Attenuation change							
Isolation decrease							
Return loss decrease							
PDL increase							
		WDL increase					

Dynamic devices		Constitution parts	Known failure mechanisms	Failure modes	Degradation acceleration factors		Relevant tests	IEC references
		Fixing point between waveguide and fibres	Dislocation by the degradation of adhesive	Insertion loss increase	Same as VOA PLC TO type			
		Fixing point of MEMS, lens and mirror	Dislocation of fixing points of optical parts	Insertion loss increase Attenuation change Isolation decrease Return loss decrease Dynamic range of attenuation decrease PDL increase WDL increase	Thermal stress High humidity (non-hermetic sealed and using adhesive) Mechanical stress	Change of temperature High temperature Damp heat Shock (storage) Vibration (storage)	IEC 61300-2-22 IEC 61300-2-18 IEC 61300-2-19 IEC 61300-2-9 IEC 61300-2-1	
		Pigtail	Same as VOA MEMS type					
Channel monitors (Performance monitors)	MEMS tuneable filters and PD type	MEMS	Stacking the moving part	Uncontrollable	Same as VOA MEMS type			
			Distortion of hinge/mirror	Wavelength deviation	Same as VOA MEMS type			
				Wavelength dynamic range change				
		Reflectance of mirror changing	PD current decrease (Error increase)	Same as VOA MEMS type				
		PD	PD degradation	PD dark current increase	Electrical surge High temperature Excess driving	ESD High temperature Maximum absolute rating test	Under study IEC 61300-2-28 Under study	
		Collimator	Dislocation of fixing points of optical parts	PD current decrease (Error increase) Return loss decrease	Same as VOA MEMS type			
		Pigtail	Same as VOA MEMS type					

Dynamic devices		Constitution parts	Known failure mechanisms	Failure modes	Degradation acceleration factors	Relevant tests	IEC references	
	Thermal tuneable filter and PD type	Thermal tuneable filter	Filter refractive index change	Wavelength deviation Wavelength dynamic range change	High temperature Excess driving	High temperature Maximum absolute rating test Driving test	IEC 61300-2-18 Under study Under study	
			Electrode degradation	Wavelength dynamic range change	Electrical surge High temperature Excess driving	ESD High temperature Maximum absolute rating test	Under study IEC 61300-2-28 Under study	
			Transmittance of tuneable filter decreasing (loss increasing)	PD current decrease (Error increase)	High temperature Excess driving	High temperature Maximum absolute rating test Driving test	IEC 61300-2-18 Under study Under study	
		PD	PD degradation	PD dark current increase	Same as channel monitor MEMS type			
		Collimator	Dislocation of fixing points of optical parts	PD current decrease (Error increase) Return loss decrease	Same as VOA MEMS type			
		Pigtail	Same as VOA MEMS type					
		Planar waveguide and PD array type	Waveguide	Refractive index changing	Insertion loss increase Tilt change Return loss decrease PDL increase WDL increase	Same as VOA PLC TO type		
	Fixing point between waveguide and fibres		Dislocation by the degradation of adhesive	Insertion loss increase	Same as VOA PLC TO type			
	PD array		PD degradation	PD dark current increase (Error increase)	Electrical surge High temperature Excess driving	ESD High temperature Maximum absolute rating test	Under study IEC 61300-2-28 Under study	

Dynamic devices		Constitution parts	Known failure mechanisms	Failure modes	Degradation acceleration factors	Relevant tests	IEC references
		PD fixing point	Dislocation of fixing points of PD array	PD current decrease (Error increase)	Thermal stress High humidity (non-hermetic sealed and using adhesive) Mechanical stress	Change of temperature High temperature Damp heat Shock (storage) Vibration (storage)	IEC 61300-2-22 IEC 61300-2-18 IEC 61300-2-19 IEC 61300-2-9 IEC 61300-2-1
		Pigtail	Same as VOA MEMS type				
Multicast optical switches	MEMS switch and fibre coupler	Optical power splitter	Tension changing of fibre fused part by deterioration of adhesive	Insertion increase PDL increase Wavelength characteristics changing TDL increase	Thermal stress High humidity	Dry heat Damp heat Change of temperature	IEC 61300-2-18 IEC 61300-2-19 IEC 61300-2-22
			Refractive index change by stress	PDL increase Wavelength characteristics changing	Thermal stress High humidity	Dry heat Damp heat Change of temperature	IEC 61300-2-18 IEC 61300-2-19 IEC 61300-2-22
			Fibre break at fused part	Insertion loss increase Return loss decrease	Mechanical stress	Vibration Shock	IEC 61300-2-2 IEC 61300-2-9
			Refractive index change by OH diffusion	Insertion loss change Wavelength characteristics changing	High humidity	Damp heat	IEC 61300-2-19
		Switching part (MEMS)	Stacking the moving part	Uncontrollable	Same as VOA MEMS type		
			Distortion of hinge/mirror	Wavelength deviation Wavelength dynamic range change	Same as VOA MEMS type		
			Reflectance of mirror changing	Insertion increase PDL increase Wavelength characteristics changing TDL increase	Same as VOA MEMS type		

Dynamic devices		Constitution parts	Known failure mechanisms	Failure modes	Degradation acceleration factors	Relevant tests	IEC references
		Collimator	Dislocation of fixing points of optical parts	Insertion increase PDL increase Wavelength characteristics changing TDL increase) Return loss decrease	Same as VOA MEMS type		
		Pigtail	Same as VOA MEMS type				
	PLC type (thermal optical effect switch)	Waveguide (splitter and thermal optical effect switch)	Refractive index changing	Insertion loss increase Crosstalk increase Return loss decrease PDL increase	Same as VOA PLC TO type		
			Electrode degradation	Crosstalk increase	Same as VOA PLC TO type		
		Fixing point between waveguide and fibres	Dislocation by the degradation of adhesive	Insertion loss increase	Same as VOA PLC TO type		
		Pigtail	Same as VOA MEMS type				
	Silicon photonics type	Waveguide (splitter and thermal optical effect switch)	Refractive index changing	Insertion loss increase Crosstalk increase Return loss decrease PDL increase	Same as VOA PLC TO type		
			Electrode degradation	Crosstalk increase	Same as VOA PLC TO type		
		Fixing point between waveguide and fibres	Dislocation by the degradation of adhesive	Insertion loss increase	Same as VOA PLC TO type		
		Pigtail	Same as VOA MEMS type				

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